

C783

Project Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A project has four activities. A (3 days) must finish before B (5 days) and C (2 days) can start. D (4 days) starts only after both B and C finish. The team's most senior engineer is assigned to B, and the sponsor wants the work done before a trade show. How much float does activity C have?

- A. 0 days, because every activity on a project is critical.
- B. 9 days
- C. 12 days
- D. 3 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An activity has an early start of day 10 and a late start of day 10. What does this tell you?

- A. It has 10 days of float.
- B. It can start any time between day 0 and day 10.
- C. It is on the critical path; any delay delays the project.
- D. It is not critical because its start dates match.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At the month-six review of a warehouse automation project, planned value is \$100,000, earned value is \$80,000, and actual cost is \$90,000. The main vendor was recently acquired by a larger firm. What is the cost performance index and what does it mean?

- A. 1.13 — the project is under budget.
- B. 0.90 — the project is over budget.
- C. 0.89 — the project is over budget.
- D. 0.80 — the project is over budget.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A project's schedule performance index is 1.2 and its planned value to date is \$50,000. What is its earned value?

- A.** \$10,000
- B.** \$60,000
- C.** \$50,000
- D.** \$41,667

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A clinic renovation has a budget at completion of \$500,000 and a CPI of 0.8. The general contractor has won two regional safety awards, and the steering committee meets monthly. If current cost performance continues, what is the estimate at completion?

- A.** \$400,000
- B.** \$125,000
- C.** \$625,000
- D.** \$500,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A software project team, including the project manager, grows from 6 people to 9 people, three of whom work remotely in another time zone. How many communication channels does the team now have?

- A.** 21
- B.** 9
- C.** 72
- D.** 36

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team has 45 communication channels. How many people are on the team?

- A.** 9
- B.** 10
- C.** 45
- D.** 23

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Midway through building a customer portal, the client's marketing lead, who represents the company's largest account, asks a developer directly to add a live-chat feature. The developer estimates it will take only two hours. What should the project manager do first?

- A. Log a change request, assess its impact, and route it through the change control process.
- B. Update the WBS and schedule to include the feature.
- C. Let the developer add it, since two hours is minor.
- D. Refuse it, because scope can never change once approved.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A construction project faces a serious risk of damage during a complex steel lift. The project manager, who has worked on eleven similar sites, hires a specialized rigging firm under a fixed-price contract that makes the firm liable for any damage. Which risk response is this?

- A. Accept
- B. Avoid
- C. Mitigate
- D. Transfer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A team finds an opportunity to finish early if it can use a partner's specialized equipment. It forms a joint arrangement so both firms split the benefit. Which response is this?

- A. Share
- B. Enhance
- C. Transfer
- D. Exploit

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A newly named project manager for a \$2 million ERP upgrade tries to schedule planning sessions, but functional managers refuse to release staff, saying the project is not official. The PM has a detailed draft schedule in a popular planning tool. Which document is missing?

- A.** A project charter signed by the sponsor.
- B.** A finished project schedule.
- C.** The project management plan.
- D.** The project scope statement.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A payroll module passed every internal test case, and the QA lead, a certified tester with ten years of experience, logged the results. Next, the client's finance director reviews the module and signs a formal acceptance form. Which process is the finance director's sign-off part of?

- A.** Close Project or Phase
- B.** Validate Scope
- C.** Control Quality
- D.** Control Scope

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A project has four activities. A (3 days) must finish before B (5 days) and C (2 days) can start. D (4 days) starts only after both B and C finish. ~~The team's most senior engineer is assigned to B, and the sponsor wants the work done before a trade show.~~ How much float does activity C have?

Target word: total float of activity C **Decides:** Path A-B-D = 12 days; path A-C-D = 9 days; C can slip $12 - 9 = 3$ days.

A	Absolute language	Only activities on the longest path have zero float.
B	Dropped element	That is the length of C's path; float needs it subtracted from the critical path.
C	Wrong stem number	That is the project duration, not C's float.
D	Correct	The critical path A-B-D is 12 days; A-C-D is 9 days, so C has 3 days of float.

2. Answer C Concept 1 • transfer

An activity has an early start of day 10 and a late start of day 10. What does this tell you?

Target word: interpret ES = LS **Decides:** Float = LS – ES = 0.

A	Wrong stem number	10 is the start day, not the float.
B	Adjacent category	Describes a window from project start, not float.
C	Correct	Zero float means the activity is critical.
D	Inverted operation	Matching dates mean zero float, which makes it critical.

3. Answer C Concept 2

At the month-six review of a warehouse automation project, planned value is \$100,000, earned value is \$80,000, and actual cost is \$90,000. ~~The main vendor was recently acquired by a larger firm.~~ What is the cost performance index and what does it mean?

Target word: CPI and interpretation **Decides:** $CPI = EV / AC = 80,000 / 90,000$.

A	Inverted operation	Divides AC by EV.
B	Wrong stem number	Divides AC by PV, which is not an earned value index.
C	Correct	$80,000 / 90,000 \approx 0.89$; below 1.0 means more is being spent than earned.
D	Confusion set	That is SPI (EV / PV), not CPI.

4. Answer B Concept 2 • transfer

A project's schedule performance index is 1.2 and its planned value to date is \$50,000. What is its earned value?

Target word: solve for EV **Decides:** $EV = SPI \times PV$.

A	Adjacent category	That is the schedule variance (EV – PV), not EV.
B	Correct	$1.2 \times \$50,000 = \$60,000$.
C	Wrong stem number	That is the planned value.
D	Inverted operation	Divides PV by SPI instead of multiplying.

5. Answer C Concept 3

A clinic renovation has a budget at completion of \$500,000 and a CPI of 0.8. ~~The general contractor has won two regional safety awards, and the steering committee meets monthly.~~ If current cost performance continues, what is the estimate at completion?

Target word: EAC at current CPI **Decides:** $EAC = BAC / CPI = 500,000 / 0.8$.

A	Inverted operation	Multiplies BAC by CPI instead of dividing.
B	Adjacent category	That is the variance at completion, not the EAC.
C	Correct	$\$500,000 / 0.8 = \$625,000$.
D	Dropped element	Ignores the cost performance to date.

6. Answer D Concept 4

A software project team, including the project manager, grows from 6 people to 9 people, ~~three of whom work remotely in another time zone.~~ How many communication channels does the team now have?

Target word: channels at $n = 9$ **Decides:** $9 \times 8 / 2 = 36$.

A	Adjacent category	That is the increase ($36 - 15$), not the total.
B	Wrong stem number	That is the number of people.
C	Dropped element	Forgets to divide by 2.
D	Correct	$9 \times 8 / 2 = 36$.

7. Answer B Concept 4 · transfer

A team has 45 communication channels. How many people are on the team?

Target word: solve for n **Decides:** $n(n - 1) / 2 = 45$, so $n(n - 1) = 90$ and $n = 10$.

A	Dropped element	Reports $n - 1$ from the formula instead of n .
B	Correct	$10 \times 9 / 2 = 45$.
C	Wrong stem number	That is the number of channels.
D	Inverted operation	Halves the channels as if $\text{people} = \text{channels} / 2$.

8. Answer A Concept 5

Midway through building a customer portal, the client's marketing lead, ~~who represents the company's largest account,~~ asks a developer directly to add a live-chat feature. ~~The developer estimates it will take only two hours.~~ What should the project manager do first?

Target word: first step for a requested change **Decides:** It changes approved scope, so it needs a change request and impact analysis.

A	Correct	Any change to the baseline is evaluated and approved before work starts.
B	Dropped element	Skips the approval step before changing baselines.
C	Camouflage	The small estimate does not remove the need for approval; this is scope creep.
D	Absolute language	Changes are allowed through formal control.

9. Answer D Concept 6

A construction project faces a serious risk of damage during a complex steel lift. The project manager, ~~who has worked on eleven similar sites,~~ hires a specialized rigging firm under a fixed-price contract that makes the firm liable for any damage. Which risk response is this?

Target word: name the threat response **Decides:** Liability moves to a third party.

A	Inverted operation	Acceptance means taking no action; here action moved the risk away.
B	Adjacent category	Avoidance changes the plan so the risk cannot occur; the lift still happens.
C	Confusion set	Mitigation reduces probability or impact but keeps the risk in-house.
D	Correct	Shifting the impact and ownership of a threat to a third party is transfer.

10. Answer A Concept 6 · transfer

A team finds an opportunity to finish early if it can use a partner's specialized equipment. It forms a joint arrangement so both firms split the benefit. Which response is this?

Target word: name the opportunity response **Decides:** An opportunity whose benefit is split with a third party.

A	Correct	Allocating an opportunity to a partner best able to capture it is sharing.
B	Adjacent category	Enhance raises probability or impact but keeps the benefit in-house.
C	Inverted operation	Transfer is the threat-side counterpart; opportunities are shared.
D	Confusion set	Exploit makes sure the opportunity happens, without a partner split.

11. Answer A Concept 7

A newly named project manager ~~for a \$2 million ERP upgrade~~ tries to schedule planning sessions, but functional managers refuse to release staff, saying the project is not official. ~~The PM has a detailed draft schedule in a popular planning tool.~~ Which document is missing?

Target word: document that authorizes the project **Decides:** No formal authorization or PM authority exists yet.

A	Correct	The charter authorizes the project and the PM's use of resources.
B	Camouflage	Answers the draft-schedule detail; a schedule does not authorize anything.
C	Adjacent category	The plan is built after authorization; it does not grant it.
D	Confusion set	Defines deliverables, not authority.

12. Answer B Concept 8

A payroll module passed every internal test case, and the QA lead, ~~a certified tester with ten years of experience,~~ logged the results. Next, the client's finance director reviews the module and signs a formal acceptance form. Which process is the finance director's sign-off part of?

Target word: process for customer acceptance **Decides:** Formal acceptance by the customer.

A	Adjacent category	Closing uses accepted deliverables; acceptance itself happens earlier.
B	Correct	Formal customer acceptance of deliverables is Validate Scope.
C	Confusion set	That was the internal testing step already completed.
D	Dropped element	Monitors scope changes but has no customer sign-off.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	6	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Critical path and float	The critical path is the longest path through the network; its activities have zero total float (float = late start – early start).	Thinking the critical path is the shortest path, or that every activity is critical.	A network of activities with durations and a question about delay or float.
2	Earned value indices	$CPI = EV / AC$ and $SPI = EV / PV$; below 1.0 means over budget or behind schedule.	Putting AC or PV on top, or mixing up CPI and SPI.	Three dollar figures labeled planned, earned, and actual.
3	Estimate at completion	If current cost performance continues, $EAC = BAC / CPI$.	Multiplying BAC by CPI, or assuming the original budget still holds.	"If current performance continues, what will the project cost?"
4	Communication channels	$Channels = n(n - 1) / 2$, where n counts everyone including the project manager.	Forgetting to divide by 2 or adding channels linearly.	A team size that changes and a question about communication complexity.
5	Integrated change control	Any change to an approved baseline goes through a change request, impact analysis, and approval before work begins.	Doing small or friendly changes informally, or refusing all changes.	A stakeholder asks a team member directly for "just a small addition."
6	Risk response strategies	Threats: avoid, transfer, mitigate, accept. Opportunities: exploit, share, enhance, accept.	Calling insurance or a fixed-price subcontract "mitigate" instead of "transfer."	Who ends up bearing the risk, and whether it is a threat or an opportunity.
7	Project charter	The sponsor-issued charter formally authorizes the project and gives the project manager authority to use organizational resources.	Thinking the project management plan or scope statement is what authorizes the project.	The PM lacks authority or the project has no formal start.
8	Validate scope vs. control quality	Control Quality checks deliverables are correct (internal); Validate Scope is the customer's formal acceptance of those deliverables.	Treating passing QA tests as customer acceptance.	"Formally accepts" or "signs off" vs. "inspects" or "tests."

Course scope (public description): Project Management is a thorough exploration of the inputs, tools, techniques, and outputs across the Project Management Body of Knowledge (PMBOK). The essential concepts and practical scenarios included enable students to build the competencies of an effective project manager. There is no prerequisite for this course.